CS445 Group Projects

A. Task Description

In this project, you are asked to implement a system for **Task 5: Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding** from SemEval. SemEval is a workshop held in conjunction with the ACL conference (a high-ranking NLP conference). Research groups release tasks with datasets, and teams compete to solve them best. You can access the train, validation, and test datasets from the Task's GitHub repository.

Unlike traditional Word Sense Disambiguation, this task acknowledges that multiple senses can be plausible in context. Using the **AmbiStory dataset** — short 5-sentence stories containing a target homonym — participants must predict human plausibility ratings (1–5) for two possible word senses, given the surrounding narrative. You can see the task details at the following link.

Task 5: Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

You will create a system to solve your chosen task using relevant literature. You can access the group sheet here.

B. Dataset

The challenge owners share their dataset. However, you can find additional datasets if you think they will help your system. You are free to curate your dataset however you like. You can also use data augmentation methods to increase your dataset.
The challenge dataset has train, validation, and test sets. While implementing and experimenting with your system, you should use only the training and validation sets. If you use the test set during training, your system will overfit, and performance for other potential test sets will be lower. You can use methods such as cross-validation to avoid overfitting. If you are going to split the dataset, you can use a predetermined seed for splitting - this ensures the same split each time you run your code.

C. Relevant Literature

To solve the task you have chosen, you need to conduct a literature search. This is the initial step of the project: to review the methodologies researchers have used to solve the problem.

- You can select one or a few papers to follow and implement them. Discuss why you have chosen these papers.
- You can also create your novel approach to solve the task, but you still have to show why you have chosen this particular approach. This requires a small discussion of the existing methods in the literature.
- Tip: You can check the task organizers' recent work. They might already be working on this specific task and might have papers out.

Suitable Venues

You are only allowed to use papers from the following venues. If you find a really good paper that is not on the list but you want to implement it, please contact Dilara Keküllüoğlu and explain your reasoning for the choice.

- SCI-Expanded list of journals
- ACL
- EMNLP
- NeurIPS
- CoNLL
- NAACL
- EACL
- COLING
- LREC

Use of Existing Codes

You can use any external libraries or open-source codes, e.g., from huggingface/kaggle, to implement your approach. If you are using open-source code, clearly show your contribution. If you use an existing code without disclosing the source or passing it off as your own code, that would count as plagiarism.

We will consider your system's performance in the evaluation; however, your contribution is more important. You should not only take some code and submit its results. You should try various approaches, including your own. **Your novel contribution to the project is more important than having better Accuracy and Spearman scores.**

D. Deliverables

- For any type of submissions, please write your group number in the submitted files.
  Ex: GroupXX_final_presentation, GroupXX_final_report, GroupXX_final_code.
- Milestone and final presentation slots will be opened in the Excel sheet, and you will be able to select your slot with your assigned evaluator. Assigned evaluators will be decided after the group formations are complete.
- You will make a presentation in your slot, but you must submit your slides along with your report and code by the deadline specified in the project submission (e.g., submit .pptx files too).
- The presentations should summarize the reports. We should be able to understand the main points of your work from the presentation. The presentation should follow the report structure for sections.
- Every team member must come to the presentation, even if they will not be the presenter.
- Please keep to the time limit. You will lose points if you go over the stated time limit for milestone and final presentations. The evaluator will stop your presentation if you go over 3 extra minutes, and we will not consider the rest of the presentation/material in their marking after this.
- Please keep to the page limit. You will lose points if you go over it.
- You should cite existing work in your slides and reports (no hallucinations).
- Please submit your code for the final deliverable in SUCourse. Please submit a clearly commented Jupyter Notebook with output logs. Do not clear the output logs!

There will be one milestone presentation and one final presentation. You need to submit reports along with the presentation slides. You will also submit your well-commented working system (Jupyter notebook and any files required to run your code) as part of the final submission.

Milestone presentations (13th April)

For the milestone, you are asked to submit a project report detailing the dataset selection. At this point, I expect you to have the dataset ready, including the extra ones you might use. You should also have a draft of your methodology plan and a list of who will do what. However, you can change your approach after the presentation if it is better for the project.

You are also asked to give a 5-minute presentation detailing your project and approaches, followed by a 5-minute Q&A (Total 10 minutes). You will schedule this with your assigned TA. You can select one person to present the milestone, but we will ask questions to everyone. These question and answers will count as your individual score. All of your team members must attend the presentation.

In the presentation,

- You need to introduce the task clearly: What is expected from the resulting system?
- What datasets are you using? Explain the task dataset and any dataset you will use for the task. How many entries are there in the dataset? What are the train/validation splits if they are given? If not, how will you split the dataset? Give details on the input/output of the requested system. Provide an EDA (Exploratory Data Analysis) on the dataset.
- Cite at least three papers you will consider in your proposed approach.
- What is your proposed approach for the task?
- Clearly divide the workload between members - the workloads should ideally be similar.

Deliverable: 5-minute presentation (with slides) + milestone report

Final presentations (18th May)

For the finals, you will submit your project report and a Jupyter notebook with your well-commented code on the 18th of May. The project report should have all the components detailed in the “Report Results” section below.

You will give a 15-minute presentation detailing your project and approaches, followed by a 5-minute Q&A (Total 20 minutes). You will schedule this with your assigned TA. Everyone will present their contribution in the project during the presentation. Aside from the group score, you will receive an individual score based on your contributions and the quality of your answers to the questions. All of your team members must attend the presentation.

The points below should be included in the presentation:

- You need to introduce the task clearly and present a small EDA. This could be a summary of the milestone work.
- You need to cite at least five papers, of which you use some in your approach. Give these citations either during the presentation or on the end slide. If not, you will not get the appropriate points. Hallucinated citations will result in deduction of marks.
- You need to explain your specific approach and clearly give the pipeline.
- If there are multiple tried approaches, you need to give explanations for all of them.
- Please clearly define what parts of the methodology you have taken from other sources (give citations) and what parts are your novel contribution, if any.
- The approach should be replicable when another person looks at your work.
- You need to give clear plots/graphs to highlight important findings. Provide confusion matrices and F1 scores, macro precision/macro recall for each approach.
- You need to compare your approach with the state-of-the-art methods and your cited papers.
- Discuss all of the points given in Section Final Reports/E.5 in the project description file.
- For individual points: Clearly outline whose contribution each work is in the presentation. We will give points for your workload and effort on the project.
- For individual points, we will ask everyone at least one question that may be related to their own contribution, the general project, or questions about their team members' contributions. You can ask for another question if you

cannot answer the first one. In that case, you will lose some points, but you can still get good enough points.

Deliverable: A 15-minute presentation (with slides) + project report + project code and files (Jupyter notebook and any files needed to run your code)

Please compress your reports, presentation slides, and project code into a single file while submitting for the final submission.

E. Report Formats

Milestone Reports

For the milestone report, please use the following outline.

1. Introduction

Give an introduction to the task you have chosen. Follow with a summary of what you have done so far.

2. Dataset Selection

What datasets are you using? Give details on why you have selected these particular datasets.

3. Approach Plan

What is the progress on the methodology decision? What are some of the main papers from which you are getting inspiration? Cite at least three of them. Have you finalized your approach, or what else will you consider to finalize your approach?

4. Next Steps

What are your next steps? How did you divide the workload between the team members? Who will do what?

Milestone reports should be no more than **four (4) pages** (excluding references and appendices) and follow the outline given above.

Final Reports

In the final project report, you are asked to have the following sections:

1. Introduction

You should give a task description. This is an introduction to the problem, the chosen methodology, and the most important results. The introduction should give a quick overview of your project.

2. Related Work

A small literature review about related work to your task: cite at least five papers that you have found. Give in-text citations. You do not need to implement all of it, but you should do a brief review of what each paper did.

3. Methodology

What dataset are you using? Which papers have you chosen to implement? Explain why you use these specific datasets and papers.
How do you train your system? Give enough detail that people can replicate your approach.

4. Results

Give the results of your system on the task. Use plots/graphs to show essential findings. Give a confusion matrix for the classification. Please give F1-score, macro precision, and macro recall. You will also give precision-recall curves. The tasks might have specialized metrics. In that case, report your results using these metrics as well. Compare it to the state-of-the-art as well as the literature you are using in your approach. Give details of the experiment. You will also write about any other methods you have tried, even if you decide not to use them in the end. Show your journey.

5. Discussion

Please discuss these points in this section:

- The selection of your dataset and its impact on system performance.
- The selection of your particular approach - what are the advantages/disadvantages of your method?
- Comparison of your results to the existing systems - discuss the results that have been reported in the Results section.
- Limitations of the proposed system.
- Potential improvements on the system if you had more time and resources.

6. Conclusion

Give another quick overview of the problem and list the most important results/discussion points to conclude the report.

7. Individual Contributions

Write a small part on who did what part in the project. We expect team members to divide the work equally. Please divide the tasks accordingly.

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The final report should be no more than **eight (8) pages** (excluding references and appendices). You can add extra results/plots to the appendix if you cannot fit them into eight pages. Make sure to include the most important results in the main report and place the extras in the appendix.

F. Grading

- 20% from the milestone presentation and report
- 35% from your final group report and code
- 5% from your evaluation results
- 40% from your individual contribution + question answers (10% from milestone + 30% from finals)

As mentioned, your novel contribution and efforts are more important than your accuracy and spearman correlation results. However, this does not mean you will not be evaluated on your results too. If your system is performing low for this task, you will receive less points. If your system performs really well, you may get bonus points. Currently, we will use the standards below for deciding whether you have an OK solution or not.

Good Solution => accuracy@std: 0.8, spearman correlation: 0.7
OK Solution => accuracy@std: 0.7, spearman correlation: 0.6
Bad Solution => accuracy@std: 0.5-0.7, spearman correlation: 0.3
Lower than this should reconsider their system and try to increase their performance..

G. AI Use

You are allowed to use AI systems to spell-check your English and improve the flow of your report.

You are **NOT** allowed to write your report from scratch using an AI system. Write the report first, and then improve on it with AI on your wording and flow.

You are **NOT** allowed to use AI systems to write your code in whole. However, you can use AI systems as detailed in the syllabus. Write your own code and detail which parts were taken from where you are using another person's code.